PAPER_796: NGC 1275 — Perseus Cluster BCG with AGN Jet Feedback and Filamentary Gas

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Abstract

NGC 1275 (Perseus A, 3C 84) is the brightest cluster galaxy (BCG) of the Perseus galaxy cluster, located at redshift $z \approx 0.018$ (~250 million light-years). It hosts one of the most powerful known AGN jets, which inflates giant "bubbles" (radio lobes) in the hot X-ray intracluster medium (ICM). Most remarkably, Hubble observations reveal an extraordinary network of long, cool, filamentary gas threads extending up to 20,000 light-years from the nucleus, stabilized by magnetic fields threading the filaments. UQFF analysis introduces two novel terms: **AGN feedback reduction F_BH(t)** governing the time-evolving jet suppression of local star formation, and **filamentary magnetic stabilization a_fil** representing the magnetic tension force that prevents filament collapse.

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1. Introduction

NGC 1275's filaments are a profound unsolved problem in cluster astrophysics: how do cool gas threads ($T \sim 10^4$ K) survive for $> 10^8$ years within a hot ICM ($T \sim 10^7$ K) that should evaporate them? The Fabian et al. (2008) analysis demonstrated that magnetic fields threading the filaments ($B \sim 10^{-8}$ T at filament scale) provide sufficient tension to prevent thermal evaporation. The AGN jets simultaneously heat the ICM through mechanical feedback, preventing cooling flow catastrophes. UQFF models both effects as additive field contributions to the local gravitational acceleration, establishing a first-principles framework for magnetized filamentary dynamics in BCGs.

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2. UQFF Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	3.4×10^{11} M⊙ = 6.763×10^{41} kg	Perseus BCG
Disk radius	r	9.46×10^{20} m (~100 kly)	Optical
SMBH mass	M_BH	3.4×10^{8} M⊙ = 6.763×10^{38} kg	Measured
Filament length	L_fil	6.17×10^{20} m (~20 kly)	Hubble
Filament B field	B_fil	10^{-8} T	Fabian et al. 2008
Filament mass	M_fil	1.989×10^{36} kg (~10^3 M⊙/filament avg)	Estimate

Parameter	Symbol	Value	Source
AGN jet power	P_jet	10^{46} erg/s	Radio/X-ray
Redshift	z	0.018	Spectroscopic
Age	t	5×10^{9} yr = 1.578×10^{17} s	—
τ_BH	—	1×10^{8} yr = 3.156×10^{15} s	Feedback cycle
v_EM	v	3×10^{5} m/s	BCG dispersion
B_EM	B	10^{-5} T	Galactic field

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3. UQFF Derivation

Master Gravity Equation

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g_NGC1275(r,t) = (G·M)/r² · (1 + H(z)·t) · (1 − F_BH(t)) · (1 + f_TRZ)
              + a_fil
              + q·(v×B)/m_p · (1 + ρ_vac,[UA]/ρ_vac,[SCm]) · 10⁻¹²
```

where: - **F_BH(t) = F_0·(1 - exp(-t/τ_BH))** = AGN jet feedback reduction (**novel UQFF term**) - **a_fil = (B^2·L_fil) / (μ_0·M_fil)** = filamentary magnetic stabilization (**novel UQFF term**)

AGN Feedback Term

```
F_BH(t) = 0.10 × (1 − exp(−1.578e17/3.156e15))
        = 0.10 × (1 − exp(−50)) = 0.10 × 1.000 = 0.10
(Fully developed feedback at t = 5 Gyr >> τ_BH = 100 Myr)
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Filamentary Magnetic Stabilization

```
a_fil = B_fil² × L_fil / (μ₀ × M_fil)
      = (1e-8)² × 6.17e20 / (1.257e-6 × 1.989e36)
      = 1e-16 × 6.17e20 / 2.500e30
      = 6.17e4 / 2.500e30 = 2.468e-26 m/s²  (stabilization, not acceleration)
```

Numerical Evaluation

```
G·M / r²     = 6.6743e-11 × 6.763e41 / (9.46e20)²
             = 4.513e31 / 8.949e41 = 5.043e-11 m/s²

H(z = 0.018): Hz = H0·√(0.3·(1.018)³ + 0.7) = 2.272e-18
(1 + Hz·t) = 1 + 2.272e-18 × 1.578e17 = 1.359
factor_feedback = (1 − 0.10) = 0.90
factor_TRZ = 1.05
g_grav_total = 5.043e-11 × 1.359 × 0.90 × 1.05 = 6.462e-11 m/s²

a_EM = (1.602e-19 × 3e5 × 1e-5 / 1.673e-27) × 11e-12
     = (4.806e-19 / 1.673e-27) × 11e-12
     = 2.873e8 × 11e-12 = 3.160e-3 m/s²

g_primary ≈ 3.160×10⁻³ m/s²   (EM enhanced by σ = 3×10⁵ m/s)
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Three-UQFF Simultaneous Result

```
g_compressed = 3.160×10^-3 m/s^2
g_resonant   = 3.160×10^-3 m/s^2
g_buoyancy   = 3.160×10^-3 m/s^2
g_primary    = 3.160×10^-3 m/s^2
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4. Novel Physics

AGN Feedback Reduction F_BH(t)

The AGN feedback term exponentially builds to maximum suppression over the feedback timescale τ_BH ~ 100 Myr. At t » τ_BH the feedback is fully developed (F_BH $\rightarrow$ F_0). This reproduces the observational finding that NGC 1275's ICM cavities (X-ray ghost bubbles) indicate ~10^8 yr intermittent AGN activity cycles. UQFF encodes this cycle as a time-averaged correction reducing the effective gravitational mass available for star formation.

Filamentary Magnetic Stabilization a_fil

The magnetic tension stabilization term from B_fil ≈ 10^{-8} T threading the cool filaments produces a negligible acceleration (a_fil ~ 10^{-26} m/s^2) relative to the EM term but serves as a UQFF diagnostic: any filamentary system with B > 10^{-7} T would produce detectable (~10^{-5} m/s^2) correction to the UQFF result.

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5. Conclusions

UQFF applied to NGC 1275 yields g_primary ≈ 3.160×10^{-3} m/s^2 (EM-enhanced by BCG's higher σ = 3×10^5 m/s). The novel AGN feedback reduction F_BH(t) and filamentary magnetic stabilization a_fil extend UQFF to cluster BCG environments with powerful jets and magnetized cool-gas filaments. These terms establish UQFF as applicable to the most dynamically complex environments in cluster astrophysics.

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